
```

clc;
clear;
close all;

% Battery parameters
initial_SOC = 100;      % Initial SOC (%)
capacity_Ah = 2;        % Battery capacity (Ah)
current_A = 1;          % Discharge current (A)
time_hours = 1;         % Discharge time (hours)

% SOC calculation using Coulomb Counting
SOC_drop = (current_A * time_hours / capacity_Ah) * 100;

final_SOC = initial_SOC - SOC_drop;

% Limit SOC between 0 and 100%
final_SOC = max(0, min(100, final_SOC));

% Display results
fprintf('Initial SOC = %.2f %%\n', initial_SOC);
fprintf('Final SOC = %.2f %%\n', final_SOC);

% Battery status
if final_SOC >= 90
    fprintf('Battery Status: High SOC\n');
elseif final_SOC >= 70
    fprintf('Battery Status: Good SOC\n');
elseif final_SOC >= 20
    fprintf('Battery Status: Medium SOC\n');
else
    fprintf('Battery Status: Low SOC\n');
end

Initial SOC = 100.00 %
Final SOC = 50.00 %
Battery Status: Medium SOC

```

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